Real-Image Reference Benchmark Report

Quantitative Concordance of Optical Flow with Manual Reference Annotations

RESEARCH BENCHMARK NOTICE: Manual annotations are reference measurements for computational benchmarking and do not represent clinical ground truth. Annotations are subject to inter- and intra-observer variability. Image-derived displacement may represent apparent image motion rather than true anatomical tissue motion. This report is strictly for computational algorithm verification.

Analysis & Reference Context

Analysis ID:	`97288ac5-601a-459b-af00-f7759da4ccd2`	**Annotation Count:**	5
Image 1:	test_bench_f0.png	**Image 2:**	test_bench_f1.png
Benchmark Score:	**72.7/100** (Research Score)	**Formula:**	Unweighted mean of Landmark Motion Concordance (1 - EPE/5.0px) and ROI Boundary IoU (IoU*100).

Paired Landmark Motion Comparison

Landmark Label	Ref Disp (px)	Est Disp (px)	Endpoint Error (px)	Disp Difference
base	4.12	4.47	1.00	0.35
apex	4.47	0.00	4.47	4.47

Manual Geometric Reference Strain

Segment	Initial L0 (px)	Final L1 (px)	Ref Strain (ε)
apex — base	50.00	51.00	0.0200

ROI & Contour Boundary Concordance

Metric	Observed Value	Interpretation
Intersection over Union (IoU)	**1.0000**	Area overlap concordance [0, 1]
Dice Coefficient	**1.0000**	Harmonic overlap measure [0, 1]
Centroid Distance	0.00 px	Distance between manual & ROI center

Benchmarking Limitations:

- Manual annotations reflect subjective operator judgment and contain intra-observer variance.
- Landmark tracking error reflects apparent intensity motion rather than physical myocardial tissue tracking.
- Engineering strain calculated from landmark chords is a 1D chord approximation and does not reflect full tensor mechanics.
- Concordance scores measure algorithmic agreement with human labels and do not establish clinical validity.